Evaluate yourself

Identify the products in the following reactions. Write their IUPAC names and mention the mechanism involved in the reactions.

i) cyclopentanol $\xrightarrow[\Delta]{H_2SO_4}$?

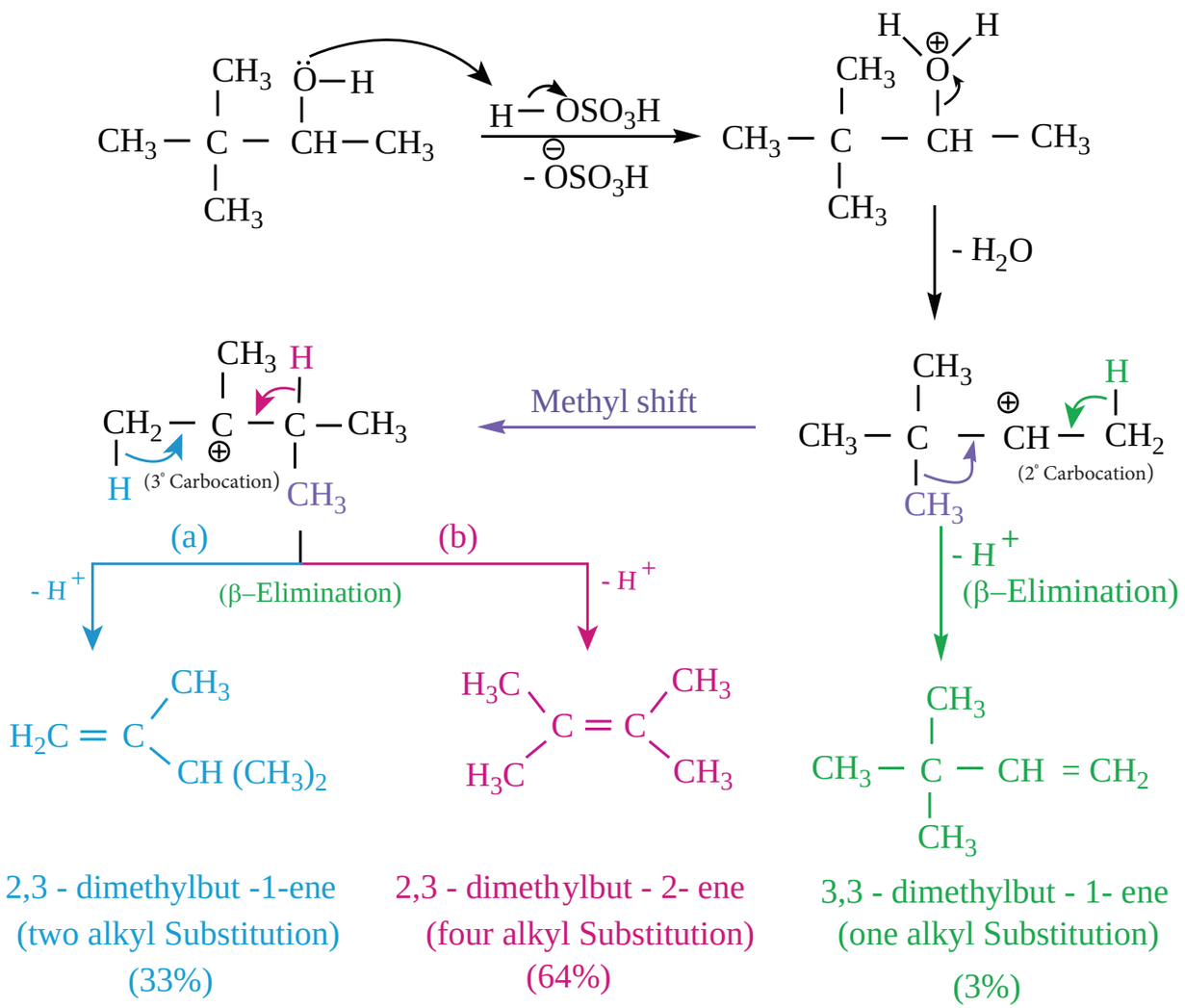
ii) butan–1–ol $\xrightarrow[H_2SO_4]{NaBr}$?

iii) neopentyl alcohol $\xrightarrow{PCl_5}$?

Saytzeff's rule

During intramolecular dehydration, if there is a possibility to form a carbon – carbon double bond at different locations, the preferred location is the one that gives the more (highly) substituted alkene i.e., the stable alkene.

For example, the dehydration of 3,3 – dimethyl – 2- butanol gives a mixture of alkenes. The secondary carbocation formed in this reaction undergoes rearrangement to form a more stable tertiary carbocation.



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